

AI Literacy: Definition, Teaching, Evaluation and Ethical Issues

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ABSTRACT

Artificial Intelligence (AI) is at the top of the agenda for education leaders today in educating the next generation across the globe. However, public understanding of AI technologies and how to define AI literacy is under-explored. This vision poses upcoming challenges for our next generation to learn about AI. On this note, an exploratory review was conducted to conceptualize the newly emerging concept “AI literacy”, in search for a sound theoretical foundation to define, teach and evaluate AI literacy. Grounded in literature on 18 existing peer-reviewed articles, this review proposed four aspects (i.e, know and understand, use, evaluate, and ethical issues) for fostering AI literacy based on the adaptation of classic literacies. This study sheds light on the consolidated definition, teaching, and ethical concerns on AI literacy, establishing the groundwork for future research such as competency development and assessment criteria on AI literacy.

KEYWORDS

AI literacy; AI learning and teaching; AI in education; AI ethics; AI literacy questionnaire.

INTRODUCTION

Artificial intelligence (AI) was first defined as “the science and engineering of making intelligent machines” in 1956 (McCarthy, 2007, p.2). Throughout several decades of the 20th century, AI has evolved progressively into intelligent machines and algorithms that can reason and adapt based on sets of rules and environment which mimic human intelligence (McCarthy, 2007). Wang (2019) broadened the definition of AI which can perform cognitive tasks particularly learning and problem-solving with the exciting technological innovations such as machine learning and neural networks (Zawacki-Richter et al., 2019). Currently, the use of AI has spread across industries (e.g., business, science, art, education) to enhance user experience and improve efficiency. Applications of AI exist in many parts of our everyday life (e.g., smart home appliances, smartphones, Google, Siri). Vast majority of the public acknowledges the existence of AI services and devices, but seldom do they know about the concepts and technology behind, or aware of potential ethical issues related to AI (Burgsteiner et al., 2016; Ghallab, 2019). Similar to classic literacy which includes reading/writing and mathematical abilities, AI literacy has emerged as a new skill set in response to this new era of intelligence. As a small number of existing articles found in the Web of Science and Scopus databases, Google Scholar search is used to identify the dramatic increase in AI literacy publications from 2014 to 2021 (see Fig. 1).

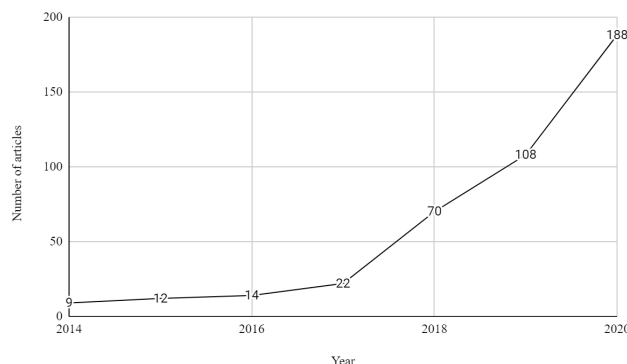


Figure 1. AI Literacy Articles from Google Scholar Published by Year

As AI becomes more and more important, researchers began to define AI literacy based on the term ‘literacy’ which has been applied to define skill sets in varied disciplines (Long & Magerto, 2020). However, few studies have provided comprehensive explanations on how to conceptualize AI literacy. To fill this gap, this study reviewed the relevant literature, and analyzed how scholars define “AI literacy”, how it can be learned, and what are the ethical concerns. Specifically, the present study poses the following four research questions: 1. How do researchers define the term “AI literacy”?; 2. How do educators help learners develop AI literacy?; 3. How do researchers evaluate students’ AI literacy skills?; 4. What are the ethical concerns in the domain of AI literacy?

METHOD

In search for literature on AI literacy, both peer-reviewed scholarly articles and conference papers from K-12 to higher education levels published from 2016 to 2021 through the *Web of Science* and *Scopus* were included in this review. The aforementioned databases were considered among the world's most trusted citation indices platforms for evidence-based quality scientific research and hence helped us to ensure the inclusion of quality scientific content (Mongeon & Paul-Hus, 2016). The articles that contained the phrase “AI literacy” in either the title, the abstract, main text or keywords were downloaded and reviewed by the researchers. The search resulted in 20 articles. After excluding irrelevant studies, as of Apr 11, 2021, a total of 18 articles were identified. Through studying the main content in the selected articles, similar meaningful concepts were identified and extracted for further thematic analysis. To establish coding reliability, six (30%) of the articles were randomly picked, blind-coded and analyzed by the two researchers. Disagreements were resolved through discussing the disputed studies.

RESULTS & DISCUSSION

RQ 1: How do researchers define the term “AI literacy”? Of the 18 articles, 12 articles defined AI literacy based on the ideas of ‘literacy’. Since the 1970s, computer applications have gained popularity across industries. It was necessary for users to become competent in using computer systems related to their specific task or job. As such, the term “digital literacy” emerged to assess basic computer-related concepts and skills. The importance of digital literacy increases as more people depend on the use of computer technologies to develop new social and economic opportunities (Leahy & Dolan, 2010). In succession to digital advancement, AI started to arise and imitate human intelligence in machines for computers to learn, reason and perceive. It was initially used in scientific research and academic environments but had yet become ubiquitous in our daily lives. Four aspects of fostering AI literacy were identified from the review (see Table 1).

AI literacy	Definitions	f	Sample references	Sample studies
Know & understand AI	Know the basic functions of AI and how to use AI applications in everyday life ethically.	18	<i>Even though transparency in algorithms and AI in general has been acknowledged to be ethically important, the public lacks understanding of even the basic functions of AI. Efforts to make AI more comprehensible exist (Robinson, 2020).</i>	Lee et al. (2021); Kandlhofer et al., 2016); Robinson (2020).
Apply AI	Applying AI knowledge, concepts and applications in different scenarios.	18	<i>Apply k-means clustering in science contexts... explore the mapping relationship between facial features and data values and apply the concept to brainstorm other objects such as Lego (Wan et al., 2020).</i>	Druga et al. (2019); Julie et al. (2020); Vazhayil et al. (2019).
Evaluate & create AI	Higher-order thinking skills (e.g., evaluate, appraise, predict, design) with AI applications.	12	<i>Design & build experiences: Technology exploration and creation activities supported students in making sense of the underlying AI concepts. (Lee, 2020).</i>	Druga et al. (2019); Han et al. (2019); How & Hung (2019).
AI ethics	Human-centered considerations (e.g., Fairness, Accountability, Transparency, Ethics)	9	<i>“AI for social good” measures an individual’s perception of the social environment surrounding the behavior, which is related to subjective norms (Chai et al., 2020).</i>	Chai et al. (2020); Druga et al. (2019); Gong et al. (2020).

Table 1. Coding Framework of AI Literacy

Know and understand AI: All 18 articles conceptualize AI literacy as educating learners about acquiring fundamental concepts, skills, knowledge and attitudes that require no prior knowledge. On top of being the end users of AI applications, learners should understand the technologies behind. Burgsteiner et al. (2016) and Kandlhofer et al. (2016) defined AI literacy as the ability to understand the basic techniques and concepts behind AI in different products and services. Many students know these AI-empowered services, but they were not familiarized with the underlying concepts such as computational thinking concepts, data structures. Sound knowledge about AI and principles of computer science will be of vast importance for their future careers (Burgsteiner et al., 2016; Kandlhofer et al., 2016). Moreover, some researchers associate AI literacy with perceived abilities, confidence and readiness in learning AI. In K-12 education, Druga et al. (2019) and Lee et al., (2021) designed learning curriculums and activities that foster AI literacy that focuses on how learners gain AI concepts.

Apply AI: All articles emphasized the importance to educate learners to know how to apply AI concepts and applications in different contexts (e.g., Druga et al., 2019; Lee et al., 2021). For example, Rodríguez-García et al. (2020) evaluated LearningML, a machine learning model builder, to educate citizens to understand AI applications and how it can affect our lives, as well as knowing the ethical issues regarding AI technologies. In addition, half of

the studies (n=9) discussed the human-centered considerations and focused on using AI concepts and application ethically, which would be further discussed in RQ4. Three out of 18 articles borrowed the ideas of computational thinking to interplay AI literacy and AI thinking (see Appendix 1). AI thinking refers to the construction of logic and algorithms in order to support students' understanding of how to use knowledge bases for problem-solving, processing semantics and handling unstructured data (Vazhayil et al., 2019). For example, How and Hung (2019) leveraged AI thinking through conducting data analytics with computing and interpreted new findings from the machine-learned discovery of hidden patterns in data.

Evaluate and create AI: AI augments human intelligence with digital automation and 12 articles alluded AI literacy to engage learners in higher-order thinking activities. Other than knowing and using AI with concepts and practices, some studies had extended AI literacy to two other competencies that enabled individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI (e.g., Long & Magerko, 2020). For example, Han et al. (2019) enhanced students' scientific and technological knowledge which then was applied in scientific research-based learning to solve practical problems. Long et al. (2019) engaged citizens in co-creating AI amenities in public spaces to broaden their public AI literacy and experiences. Students with the ability to evaluate and create AI could infer from, connect, manipulate, and categorize AI concepts together in novel ways. Overall, although these articles showed slight variations on the definition of AI literacy, they support the notion that everyone, especially K-12 children, acquire basic AI knowledge and abilities, and enhance motivation and career interest (Chai et al., 2020b). In addition to knowing and using AI ethically, AI literacy serves as a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI (Long & Magerko, 2020).

RQ 2. How do educators teach and learn AI? In K-12 education, researchers (n=12) designed learning curriculums and activities that foster AI literacy that focuses on how learners gain AI concepts, as well as using AI to incorporate into their own applications (e.g., Druga et al., 2019; Lee et al., 2021). Long and Magerko (2020) and Rodríguez-García et al. (2020) mentioned Touretzky et al (2019)'s five "big ideas" of AI which provide a strong foundation for future research on fostering AI literacy (see Appendix 2). In another study, Rodríguez-García et al. (2020) evaluated LearningML, a machine learning model builder, to develop critical thinking, and K-12 children on AI fundamentals to understand what can be made with AI, how it can affect their lives, and the ethical issues regarding AI technologies. In higher education, two researchers claimed that AI knowledge and skills become more advanced to meet the future job demands. Kandlhofer et al. (2016) and Burgsteiner et al. (2016) listed a set of AI concepts that has potential to become the basis for careers in science and engineering: automata, intelligent agents, graphs and data structures, basics of computer science, machine learning, etc. based on the AI bible written by Russell and Norvig (2009). Four studies mentioned the importance of educating citizens on fundamental AI concepts, and the impacts of AI technologies on their everyday lives. For example, Robinson (2020) mentioned that the Norwegian policy document titled "AI for everyone: Elements of AI" (p. 44) asserts the government will make AI learning courses globally accessible in 2020, which conceptualizes AI literacy as educating their citizens about the elements of AI that require no prior knowledge (Robinson, 2020).

Learning artefacts: Given the complexity of AI, age-appropriate learning artefacts were important to scaffold students' AI conceptual understandings, and enhance their motivation and interest in learning AI. In recent years, there has been an increase in hardware and software that enhance AI concepts accessible to younger learners. Appendix 3 provides an overview of the types of AI learning artefacts ranging from hardware (n=5) to software-focused artefacts (n=3), intelligent agents (n=4) and unplugged learning tools (n=5). The democratization of current AI technologies encourages students to make intelligent agents and machine learning models without needing to program (Long & Magerko, 2020). In this context, we can see an opportunity for educators to democratize access to AI literacy and reinforce the AI concepts through these emerging tools. Alternatively, educators designed unplugged learning activities to foster students' AI literacy without using a computer through engaging approaches such as case study, role-playing and storytelling (e.g., Julie et al., 2020).

RQ 3. How do researchers evaluate students' AI literacy skills? Among 18 studies, 11 articles adopted quantitative (n=9) and qualitative (n=6) evaluation methods to examine how students enhance their literacy skills (see Appendix 4). To evaluate K-12 students' AI literacy, one important component is to promote their intention to learn and possess basic knowledge about AI. Nine studies assessed the knowledge acquisition of K-12 and university students via group discussion and pre- and post-knowledge tests (e.g., What are the characteristics of depth-first search?), and students' perceived abilities (e.g., How would you rate your knowledge about search algorithms?) (Kandlhofer et al., 2016; Wan et al., 2020). Other than self-reported questionnaires, the study assessed students' output that used artefacts such as computer programs, documentations and presentations of their experiments. Five researchers also collected qualitative data by taking pictures, field notes during teaching, and interviewing students to understand their motivations, expectations and lessons learned. Druga (2019) recorded students' interaction with AI agents through field observations and adopted a three-attribute AI perception questionnaire to evaluate how 102 children (7-12 years old) interacted and perceived their AI agents in their lessons.

These 3 attributes measure whether the agents are smarter, truthful and understand them. Three studies discussed other aspects such as confidence in using AI, motivation, AI for societal good. In another study, Chai et al. (2020a) measured AI literacy as students' perception of their knowledge and skills of using AI technology, and designed and administered an instrument to 545 students (aged 13-18) with 7 aspects. Dai et al. (2020) further included 'relevance' as the eighth aspect that is associated with AI literacy. They considered AI literacy as the knowledge foundation that provides students with a perception of AI technology.

RQ 4. What are the ethical concerns in the domain of AI literacy? As AI plays an important role in day-to-day decision making, misused or poorly designed AI could cause irreparable harm to humans and the society (Fourtané, 2020). AI-concerned scientists and engineers like Elon Musk expound on the horrors that future AI technologies may wreak on humanity in decades to come (Johnson, 2019). However, only half of the studies in this review had mentioned human-centered considerations and raised attention to educate citizens to become socially responsible (Ahmad et al., 2020). Gong et al. (2020) found that students pay little attention to ethical concerns such as bias in AI and legal responsibility (8%), and intellectual property (9%). As such, educators should not only focus on enhancing students' AI abilities and interests, but also help students to realize the societal impact and ethical concerns.

To bring up future responsible citizens who compete in using AI in a reliable, trustworthy and fair way, computational techniques must be facilitated in an innovative and responsible manner, while prioritizing issues of fairness, accountability, transparency, and ethics by drawing on fields with a sociotechnical orientation (Hagendorff, 2021; Microsoft, 2021). AI literacy in K-12 education is at its starting point. It highlights how people comprehend AI concepts and apply AI. Broadening participation in AI for all is necessary to ensure that the design and utilization of AI technologies are inclusive to address under-representation of people of colour and women in AI. For example, Lee et al. (2021) designed a middle-school AI literacy curriculum on ethics education that incorporates algorithmic bias, ethical design of recommender systems, unanticipated consequences into a series of AI learning activities such as training models with Teachable Machine, generating texts and redesigning videos. A student gave a feedback during the interview: "If people who make the AIs don't think enough through it, they can have bad consequences. They will get sued if they don't check over or get into damages" (p.194). The study envisioned that the foundation of future AI industries would be built on "principles of inclusivity, provide equitable access, include consideration of multiple stakeholders and potential users, and minimize the potential for bias" (p.191). Another study conducted by Druga (2019) engaged 108 children (aged 7-12) from different countries and social economics status to construct a guideline to discuss how educators can design inclusive learning activities such as avoiding deceiving technologies, offering ways for children to customize and program the machine, and encouraging reflection and collaboration by allowing children to share and modify each other's projects. Moreover, Robinson (2020) conducted a document analysis and pointed out that a guiding framework for AI government policy is important for citizens to have a common ethical, responsible and human-centered basis to better support the development of AI technology in their society. With the democratization of current AI technologies, educators should not only teach students to build machine learning models, but also guide them on how to implement these emerging technologies ethically. To summarize, conceptualizing AI literacy with human-centered considerations is crucial to building a future inclusive society.

CONCLUSION

In this review, a variety of definitions of AI literacy was noticed. The most common approach to define AI literacy is to base on different types of 'literacies' which have recently been applied to define skill sets in varied disciplines. In our review, most researchers advocated that instead of merely knowing how to use AI applications, learners should be inculcated with the underlying AI concepts for their future career, as well as the ethical concerns of AI applications to become a responsible citizen. Since AI literacy is an emerging field that there is a lack of journals published in this field, several limitations are found in this review. The keyword search limits the scope of domain specificity within the AI context without focusing on other subfields of AI like machine learning, neural network, etc. Some authors designed interventions and learning programs to discuss how to foster AI literacy in their studies without explicitly defining it. Although we believe there is a larger pool of studies concerning the development of AI literacy, they may not be entirely captured to avoid any misinterpretation of their underlying concepts in AI literacy. In addition, future studies are needed to examine effective means to foster students' AI literacy, its assessment criteria and ethical concerns. We hope this review will inspire scholars, educators, and government officers to begin the discussion on how to define, implement and evaluate AI literacy in the future.

REFERENCES

- Ahmad, M. A., Teredesai, A., & Eckert, C. (2020, January). Fairness, accountability, transparency in AI at scale: Lessons from national programs. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 690-690).
- Baker, T., Smith, L., and Anissa, N. (2019). *Educ-AI-Tion Rebooted? Exploring the Future of Artificial Intelligence in Schools and Colleges* (London: Nesta). Available online at: <https://www.nesta.org.uk/report/education-rebooted>.

- Burgsteiner, H., Kandlhofer, M., & Steinbauer, G. (2016, March). Irobot: Teaching the basics of artificial intelligence in high schools. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 30, No. 1).
- Chai, C. S., Wang, X., & Xu, C. (2020a). An extended theory of planned behavior for the modelling of Chinese secondary school students' intention to learn Artificial Intelligence. *Mathematics*, 8(11), 2089.
- Chai, C. S., Lin, P. Y., Jong, M. S. Y., Dai, Y., Chiu, T. K., & Huang, B. (2020b, August). Factors Influencing Students' Behavioral Intention to Continue Artificial Intelligence Learning. In *2020 International Symposium on Educational Technology (ISET)* (pp. 147-150). IEEE.
- Dai, Y., Chai, C. S., Lin, P. Y., Jong, M. S. Y., Guo, Y., & Qin, J. (2020). Promoting students' well-being by developing their readiness for the artificial intelligence age. *Sustainability*, 12(16), 6597.
- Druga, S., Vu, S. T., Likhith, E., & Qiu, T. (2019). Inclusive AI literacy for kids around the world. In *Proceedings of FabLearn 2019* (pp. 104-111).
- Ertmer, P. A., & Ottenbreit-Leftwich, A. T. (2010). Teacher technology change: How knowledge, confidence, beliefs, and culture intersect. *Journal of research on Technology in Education*, 42(3), 255-284.
- Fourtané, S. (2020, August). Ethics of AI: Benefits and Risks of Artificial Intelligence Systems. *Interesting Engineering*. Retrieved from <https://interestingengineering.com/ethics-of-ai-benefits-and-risks-of-artificial-intelligence-systems>
- Genlott, A. A., & Grönlund, Å. (2013). Improving literacy skills through learning reading by writing: The iWTR method presented and tested. *Computers & education*, 67, 98-104.
- Ghallab, M. (2019). Responsible AI: requirements and challenges. *AI Perspectives*, 1(1), 1-7.
- Gong, X., Tang, Y., Liu, X., Jing, S., Cui, W., Liang, J., & Wang, F. Y. (2020, October). K-9 Artificial Intelligence Education in Qingdao: Issues, Challenges and Suggestions. In *2020 IEEE International Conference on Networking, Sensing and Control (ICNSC)* (pp. 1-6). IEEE.
- Hagendorff, T. (2020). The ethics of AI ethics: An evaluation of guidelines. *Minds and Machines*, 30(1), 99-120.
- Han, X., Hu, F., Xiong, G., Liu, X., Gong, X., Niu, X., ... & Wang, X. (2018). Design of AI+ Curriculum for Primary and Secondary Schools in Qingdao. In *2018 Chinese Automation Congress (CAC)* (pp. 4135-4140). IEEE.
- How, M. L., & Hung, W. L. D. (2019). Educating AI-thinking in science, technology, engineering, arts, and mathematics (STEAM) education. *Education Sciences*, 9(3), 184.
- Johnson, K (2019, November). AI ethics is all about power. *Venturebeat Insider - The Machine Making Sense of AI*. Retrieved from <https://venturebeat.com/2019/11/11/ai-ethics-is-all-about-power/>
- Julie, H., Alyson, H., & Anne-Sophie, C. (2020, October). Designing digital literacy activities: an interdisciplinary and collaborative approach. In *2020 IEEE Frontiers in Education Conference (FIE)* (pp. 1-5). IEEE.
- Kandlhofer, M., Steinbauer, G., Hirschmugl-Gaisch, S., & Huber, P. (2016, October). Artificial intelligence and computer science in education: From kindergarten to university. In *2016 IEEE Frontiers in Education Conference (FIE)* (pp. 1-9). IEEE.
- Leahy, D., & Dolan, D. (2010, September). Digital literacy: A vital competence for 2010?. In *IFIP International Conference on Key Competencies in the Knowledge Society* (pp. 210-221). Springer, Berlin, Heidelberg.
- Long, D., & Magerko, B. (2020, April). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1-16).
- Long, D., Jacob, M., & Magerko, B. (2019). Designing co-creative AI for public spaces. In *Proceedings of the 2019 on Creativity and Cognition* (pp. 271-284).
- McCarthy, J. (2007). From here to human-level AI. *Artificial Intelligence*, 171(18), 1174-1182.
- McClelland, C. (2020). The Impact of Artificial Intelligence - Widespread Job Losses. Retrieved from <https://www.iotforall.com/impact-of-artificial-intelligence-job-losses>
- Microsoft. (2021). FATE: Fairness, Accountability, Transparency, and Ethics in AI. Retrieved from <https://www.microsoft.com/en-us/research/theme/fate/>
- Mohammadyari, S., & Singh, H. (2015). Understanding the effect of e-learning on individual performance: The role of digital literacy. *Computers & Education*, 82, 11-25.
- Mongeon, P., & Paul-Hus, A. (2016). The journal coverage of Web of Science and Scopus: a comparative analysis. *Scientometrics*, 106(1), 213-228.
- Ng, T. K. (2021). [New interpretation of extracurricular activities via social networking sites: A case study of artificial intelligence learning at a secondary school in Hong Kong](#). *Journal of Education and Training Studies*, 9(1), 49-60.
- Ng, T.K. & Chu, S. K.W. (2021). Motivating students to learn AI through social networking sites: A case study in Hong Kong. *Online Learning*, 25(1), 195-208.
- Rodríguez-García, J. D., Moreno-León, J., Román-González, M., & Robles, G. (2020, October). Introducing Artificial Intelligence Fundamentals with LearningML: Artificial Intelligence made easy. In *Eighth International Conference on Technological Ecosystems for Enhancing Multiculturality* (pp. 18-20).

- Russell Stuart, J., & Norvig, P. (2009). *Artificial intelligence: a modern approach*. Prentice Hall.
- Schaper, M. M., Malinverni, L., & Valero, C. (2020, October). Robot Presidents: Who should rule the world? Teaching Critical Thinking in AI through Reflections upon Food Traditions. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* (pp. 1-4).
- Samuels, A. (2020). Millions of Americans Have Lost Jobs in the Pandemic—And Robots and AI Are Replacing Them Faster Than Ever. Retrieved from <https://time.com/5876604/machines-jobs-coronavirus/>
- Vazhayil, A., Shetty, R., Bhavani, R. R., & Akshay, N. (2019, December). Focusing on teacher education to introduce ai in schools: Perspectives and illustrative findings. In *2019 IEEE Tenth International Conference on Technology for Education (T4E)* (pp. 71-77). IEEE.
- Wang, P. (2019). On defining artificial intelligence. *Journal of Artificial General Intelligence*, 10(2), 1-37.
- Zawacki-Richter, O., Marín, V. I., Bond, M., and Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education – where are the educators? *Int. J. Educ. Technol. Higher Edu.* 16:39. doi: 10.1186/s41239-019-0171-0

APPENDIX 1–5

For more details, please visit <https://bit.ly/3dsiJ4g>.